

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Java, JavaScript, TypeScript, C++, SQL, Bash

AI Agents & LLMs

AI agents, agentic workflows, tool use, prompt design, LLM evaluation, OpenAI / Groq APIs, multi-provider routing

RAG & Retrieval

RAG, knowledge-graph RAG, multi-view retrieval, RRF, cross-encoder reranking, MMR, sentence-transformers

Open-Source Models

Llama (3 / 7B), llama.cpp, 4-bit quantization, Ollama, HuggingFace Transformers, PyTorch, TensorFlow

Integration & Cloud

FastAPI, Node.js, REST APIs, Docker, AWS, GCP, Azure, Git, CI/CD, system integration

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.
- Worked **cross-functionally** with clinical, product, and AI teams to integrate features into existing production systems and translate requirements for **non-technical stakeholders**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product for context-aware replies (prompt design, response-quality heuristics), and shipped an **open-source LLM** feature in production, 4-bit quantized **Llama-7B via llama.cpp** with persona prompts and a content-safety filter, integrated into the live product end-to-end with no ML mentor.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables using typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

KG2RAG-Enhanced - Retrieval-Augmented Generation

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama + LLaMA-3, sentence-transformers, PyTorch | UMass CS 685 (team)

- Led the **multi-view retrieval** component of a team **RAG** pipeline (sub-question decomposition, per-view retrieval, RRF fusion + cross-encoder reranking + MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.
- Ran the pipeline on an **open-source** Ollama + LLaMA-3 stack with token/latency logging, improving the relevance and accuracy of generated answers, the core goal of a RAG system.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an edge-native AI research assistant: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step tasks survive process restarts and tool failures, with conversation state persisted in D1.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.